

Zachary Espinosa

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EDUCATION

University of Washington , Seattle, WA PhD, Atmospheric and Climate Science Data Science Specialty	Jun 2026
Stanford University , Stanford, CA M.S. Applied and Engineering Physics, Schools of Arts and Science	Jun 2021
Stanford University , Stanford, CA B.S. Computer Science, School of Engineering Concentration: Artificial Intelligence	Sep 2020

HONORS & FELLOWSHIPS

Department of Energy Computational Science Graduate Fellowship (DOE CSGF)	Apr 2022
Graduate Student Equity & Excellence Fellowship (GSEE Fellow)	Sep 2021
Achievement Rewards for College Scientists Foundation Scholar (ARCS Scholar)	Sep 2021
The GEM National Consortium Graduate Fellow (GEM Graduate Fellow)	Jan 2020

PUBLICATIONS

- [11] **Espinosa, Z.**, Cresswell-Clay, N., Yik, W., Bitz, C., Blanchard-Wrigglesworth, E., Harrington, P., Pruitt, D., Pritchard, M., & Durran, D. "DLESyM-Ocean: A Deep Learning Probabilistic Global Model for Simulating Present-Day Upper Ocean and Sea Ice." arXiv preprint (2026): <https://doi.org/10.48550/arXiv.2608.11545> [submitted]
- [10] Li Q., **Espinosa, Z.**, Armour, K., Cheng W., & Thompson L., "The Double-ITCZ Bias Weakens the Southern Ocean-Tropical Pacific Teleconnection." ESS Open Archive (2026). <https://doi.org/10.22541/essoar.15003562/v1> [submitted]
- [9] **Espinosa, Z.**, Zelinka M., Bitz, C., & Armour, K. "The Southern Ocean and Tropical Eastern Pacific Teleconnection in models and observations". *Authorea Preprint* (2025): [10.22541/essoar.174534450.08930960/v1](https://doi.org/10.22541/essoar.174534450.08930960/v1)
- [8] Clemens-Sewall, D., **Espinosa, Z.**, Oggier, M., Sturm, M., Webster, M., Wesen, S., ... & Delamere, J. "Snow ALbedo eVolution (SALVO) Campaign Measurements from April-June, 2024 in Utqiagvik, AK." Atmospheric Radiation Measurement User Facility (2025). [[Spectral Albedo](#)] [[Oblique Photos](#)] [[Snow Depth](#)] [[Broadband Albedo](#)]
- [7] **Espinosa, Z.**, Roach, L., Bitz, C., & Notz, D. "Sea Ice in Earth system models". *Sea Ice 4* (2025): [10.1002/9781394213764.ch10](https://doi.org/10.1002/9781394213764.ch10)
- [6] **Espinosa, Z.**, Blanchard-Wrigglesworth, E., and Bitz, C. "Understanding the drivers and predictability of record low Antarctic sea ice in austral winter 2023." *Nature Communications Earth & Environment* (2024): [10.1038/s43247-024-01772-2](https://doi.org/10.1038/s43247-024-01772-2)
- [5] Cresswell-Clay, N., Liu, B., Durran, D., Liu, A., **Espinosa, Z.**, Moreno, R., & Karlbauer, M. "A Deep Learning Earth System Model for Stable and Efficient Simulation of the Current Climate". *AGU Advances* (2025): [10.1029/2025AV001706](https://doi.org/10.1029/2025AV001706)
- [4] Schneider, D. P., Yin, Z., O'Connor, G. K., Blanchard-Wrigglesworth, E., Cast, Z. I., Datta, R., & **Espinosa, Z.** "Increasing Antarctic snowfall mitigates sea level rise less than projected due to meltwater influence on sea surface temperatures". *Authorea Preprint* (2024): [10.22541/essoar.172411232.25724214/v1](https://doi.org/10.22541/essoar.172411232.25724214/v1)
- [3] **Espinosa, Z.**, and Zelinka, M. "The shortwave cloud-SST feedback amplifies multi-decadal Pacific sea surface temperature trends: Implications for observed cooling." *Geophysical Research Letters* 51.18 (2024): e2024GL111039. [10.1029/2024GL111039](https://doi.org/10.1029/2024GL111039)
- [2] Blanchard-Wrigglesworth, E., Cox, T., **Espinosa, Z.**, & Donohoe, A. "The largest ever recorded heatwave—Characteristics and attribution of the Antarctic heatwave of March 2022". *Geophysical Research Letters*, 50(17) (2023): e2023GL104910. [10.1029/2023GL104910](https://doi.org/10.1029/2023GL104910)
- [1] **Espinosa, Z.**, et al. "Machine learning gravity wave parameterization generalizes to capture the QBO and response to increased CO₂." *Geophysical Research Letters* 49.8 (2022): e2022GL098174. [10.1029/2022GL098174](https://doi.org/10.1029/2022GL098174)

PROFESSIONAL EXPERIENCE

Research Scientist <i>San Francisco, CA</i> Google DeepMind AI for weather forecasting	May 2025 – Present
Student Researcher <i>San Francisco, CA</i> Google DeepMind AI for high-resolution, regional weather forecasting	Aug 2025 – Dec 2025

Technical Intern, PhD <i>San Francisco, CA</i> Brightband	May 2025 – Jul 2025
• Diffusion modeling for AI data assimilation for weather forecasting	
Research Intern, PhD <i>Livermore, CA</i> Lawrence Livermore National Laboratory	Jun 2023 – Sep 2023
• Studied the impact of marine boundary layer clouds on historical East Pacific Ocean cooling	
Research Intern, PhD <i>Richland, WA</i> Pacific Northwest National Laboratory	Jun 2021 – Sep 2021
• Studied the impact of climate change on annual precipitation in the Amazon Rainforest	
Graduate Research Assistant <i>Stanford, CA</i> Stanford Earth Systems Science	Sep 2019 – Sep 2021
• Developed a machine learning parameterization of gravity wave in a global climate model (Sheshadri Group)	
• Publication in Geophysical Research Letters - Espinosa, Zachary I., et al (2022)	
Machine Learning Engineering Intern <i>Redwood City, CA</i> UnifyID	Apr 2020 – Jun 2020
• Developed in-house machine learning pipeline for research & development. Introduced pipeline testing	
Quantum Engineering Intern <i>Palo Alto, CA</i> AT&T Foundry	Jun 2019 – Sep 2019
• Built an open-source python framework for quantum networking (QN) simulations called netQuil , designed to support the implementation of canonical QN protocol (e.g. teleportation, superdense coding)	
Software Engineering Intern <i>Mountain View, CA</i> Smartcar, Inc.	Jan 2019 – Jun 2019
• Designed, built, and launched electric vehicle endpoints for Smartcar API	
• Maintained python, node.js, and java SDKs. Contributed to OAuth2 pipeline.	
Mobile Software Engineering Intern <i>San Francisco, CA</i> OXO, Inc.	Apr 2018 – Sep 2018
• Built first iteration MVP mobile app for iOS and Android using React Native, Firebase, Heroku, and AWS RDS.	
Web and Networking Engineering Intern <i>Ashton, ID</i> Henry's Fork Foundation	Jun 2017 – Sep 2017
• Designed and built a data collection network for monitoring the Yellowstone watershed.	

TEACHING, MENTORSHIP & SERVICE

Instructor & Mentor AI Fellowship Program <i>VeritasAI</i>	Jun 2024 – Present
Graduate Student Representative UW Program on Climate Change <i>Seattle, WA</i>	Sep 2022 – Sep 2024
Graduate President of UW American Meteorological Society Chapter <i>Seattle, WA</i>	Sep 2021 – Sep 2023
Guest Lecturer ATMS 220: Exploring the Atmospheric Sciences <i>Seattle, WA</i>	Oct 2023
Guest Lecturer ATMS 220: Exploring the Atmospheric Sciences <i>Seattle, WA</i>	May 2023
Teaching Assistant ATMS 101: Weather <i>Seattle, WA</i>	Jan 2023 – Mar 2023

PRESENTATIONS

Talk AMS 2025 Seasonal Sea Ice Forecasting with a Deep Learning Earth System Model	Jan 2025
Talk AMS 2025 The Impact of the Shortwave Cloud Feedback on East Pacific Multi-Decadal Variability	Jan 2025
Poster AMS 2025 Record Low Antarctic Sea Ice in Austral Winter 2023: Mechanisms and Predictability	Jan 2025
Talk Catalyst Project The Shortwave Cloud-SST Feedback Amplifies Multi-Decadal Pacific SST Trends	Nov 2024
Poster CFMIP 2024 The Impact of the Shortwave Cloud Feedback on East Pacific Multi-Decadal Variability	Jun 2024
Talk UW Climate Dynamics Seminar From Record Low Sea Ice to East Pacific Cooling: Unraveling SH Extremes	Apr 2024
Poster US CLIVAR Blocking and Extreme Weather Workshop The Physics of Antarctic Heatwaves	Mar 2024
Poster AGU Fall Meeting The Physics of Antarctic Heatwaves	Dec 2023
Poster Graduate Climate conference Drivers of Record Low Antarctic Sea Ice in Austral Winter 2023	Oct 2023
Poster DOE CSGF Annual Review The Physics of Antarctic Heatwaves	Jul 2023
Talk Scientific Committee on Antarctic Research The Physics of Summertime Antarctic Heatwaves	June 2023
Poster BEPSII Arctic Field School Drivers of Interannual Variability of Summer Sea Ice Extent	May 2022
Talk AGU Fall Meeting Machine Learning Emulation of Parameterized Gravity Wave Momentum	Dec 2021
Talk EGU General Assembly Machine Learning Emulation of Parameterized Gravity Wave Momentum	Apr 2021
Talk CalGFD A Data-Drive, Single column Gravity Wave Parameterization in an Idealized Model	Aug 2020
Poster Stanford Deep Learning Poster Session Distracted Driver Detection	Jun 2018
Poster Stanford Artificial Intelligence Post Session Tracking Schistosomiasis with Computer Vision	Mar 2018

ADDITIONAL INFORMATION

Tooling: Python, PyTorch, High Performance Computing, multi-node & multi-gpu training, Dask, Fortran, C, C++

Expertise: Generative AI, Diffusion Models, ConvNets, Vision Transformers, AI Autoregressive Weather & Climate Prediction, Hybrid and Full AI Climate modeling, Climate Dynamics, Extreme Weather, Subseasonal to Seasonal Forecasting, Data Analytics